

=====

ROW_NUMBER
RANK
DENSE_RANK

```
SELECT *,  
ROW_NUMBER() OVER (ORDER BY hire_date) as row_num,  
RANK() OVER (ORDER BY hire_date) as rnk,  
DENSE_RANK() OVER (ORDER BY hire_date) as dense_rnk  
FROM EMPLOYEES;
```

-- when there are no ties/duplicates then all of them will give
-- the same result.

the difference comes when there are ties

```
SELECT *,  
ROW_NUMBER() OVER (ORDER BY salary desc) as row_num,  
RANK() OVER (ORDER BY salary desc) as rnk,  
DENSE_RANK() OVER (ORDER BY salary desc) as dense_rnk  
FROM EMPLOYEES;
```

-- 2 people with same value have the same rank and same dense rank

-- 2 people with same value have the same rank and will skip the next rank

RANK()

-- dense rank is similar to rank but this will not skip anything

DENSE_RANK()

-- rank leaves gaps in ranking for tied data.

-- while row_number assigns a unique number to each row without
-- considering the ties.

-- dense_Rank assigns identical ranks to tied data, without any gaps.

-- top 10 customers with max purchase from each city

-- top 3 products with max sales from each city.

-- 3 newest employees from each department

-- top selling product from each category

-- top 3 students in each subject